Five-Stock Portfolio Optimization & Out-of-Sample Strategy Review

AAPL · JPM · JNJ · XOM · WMT | 60 monthly returns | SPY benchmark

Investment research report | Analysis based on APPLE(10).xlsx | 9 November 2026

Executive summary

The central result is not that optimization consistently beat a simple allocation. It did not. Over the 12-month out-of-sample period, monthly reoptimization improved the Minimum Variance portfolio but materially weakened the Maximum Sharpe strategy. A monthly rebalanced Equal-Weighted portfolio provided the strongest realized risk-adjusted outcome, despite trailing rolling Minimum Variance in absolute return.

Rolling Minimum Variance earned 37.20% gross (36.89% net of modeled transaction costs), versus 32.27% gross (32.07% net) for Equal-Weighted. Yet Equal-Weighted experienced substantially less volatility (9.52% gross versus 12.86%) and a smaller maximum drawdown (1.91% versus 4.39%). Rolling Maximum Sharpe earned 20.19% gross and fell as much as 10.86% from its prior peak. The workbook reports a provisional 10.81% SPY return; its final October 2026 observation is derived from an assumed −0.88% monthly return and requires independent verification.

Recommendation: prefer the Equal-Weighted allocation when the mandate emphasizes risk-adjusted returns, drawdown control and a transparent implementation. Rolling Minimum Variance merits consideration where higher absolute returns justify greater realized risk and trading activity. This is a retrospective finding from one 12-month period, not evidence of durable future superiority.

1 Research question and investment universe

This study asks whether mean–variance optimization improves realized outcomes relative to a straightforward 20% allocation to each security, and whether monthly reoptimization adds value compared with holding weights selected once. The investment universe comprises Apple (AAPL), JPMorgan Chase (JPM), Johnson & Johnson (JNJ), Exxon Mobil (XOM) and Walmart (WMT). SPY serves as the intended broad US equity-market reference, subject to the price-data verification described in Section 7.

The five-stock universe spans technology, banking, healthcare, energy and consumer retail. That sector mix introduces distinct business exposures, but it is not equivalent to broad market diversification: idiosyncratic company risk remains meaningful, and the asset selection itself was not optimized in the model.

2 Data and portfolio construction

The workbook contains 60 monthly stock-return observations. The first 48 months form the estimation sample and the final 12 months form the out-of-sample evaluation. For the static portfolios, weights are estimated once using the initial 48 months. For the rolling portfolios, a 48-month estimation window advances by one month before each subsequent test-month allocation. This preserves the training-before-testing sequence.

All optimized portfolios are fully invested and long-only, with individual weights bounded between 0% and 100%. Expected annual returns are estimated as 12 times average monthly return; annualized volatility uses monthly covariance multiplied by 12. The Sharpe ratio uses a fixed 4% annual risk-free assumption. Equal-Weighted is restored to 20% in each name each month, which is why its static and rolling series coincide in the workbook.

The study begins with a hypothetical USD 1,000,000 allocation in its portfolio construction worksheets and uses USD 100,000 to illustrate out-of-sample wealth growth. Performance percentages are independent of this display scale.

Initial full-sample allocation context

Security	Equal-weight	Minimum Variance	Maximum Sharpe
AAPL	20.00%	19.07%	25.48%
JPM	20.00%	11.68%	6.90%
JNJ	20.00%	38.83%	3.18%
XOM	20.00%	10.45%	30.95%
WMT	20.00%	19.98%	33.50%

These are the allocations displayed in the standalone optimization worksheets, which use a different estimation scope from the 48-month backtest. They illustrate the optimizer's preferred exposures but should not be presented as the backtested starting weights. The backtest uses its own training-sample allocations.

3 Static out-of-sample performance

Portfolio	12m return	Ann. volatility	Sharpe	Max drawdown
Equal-Weighted	32.27%	9.52%	2.60	−1.91%
Minimum Variance	35.70%	13.44%	2.06	−5.26%
Maximum Sharpe	29.40%	15.93%	1.46	−8.34%
SPY	10.81%*	13.51%	0.53	−5.96%

The static Minimum Variance portfolio delivered the highest absolute return among the fixed-weight strategies, despite being constructed to minimize estimated volatility rather than maximize returns. Equal-Weighted nevertheless led on realized Sharpe ratio and drawdown. The static Maximum Sharpe portfolio underperformed both alternatives on these out-of-sample measures, showing that a strong training-period objective does not necessarily translate into superior subsequent performance.

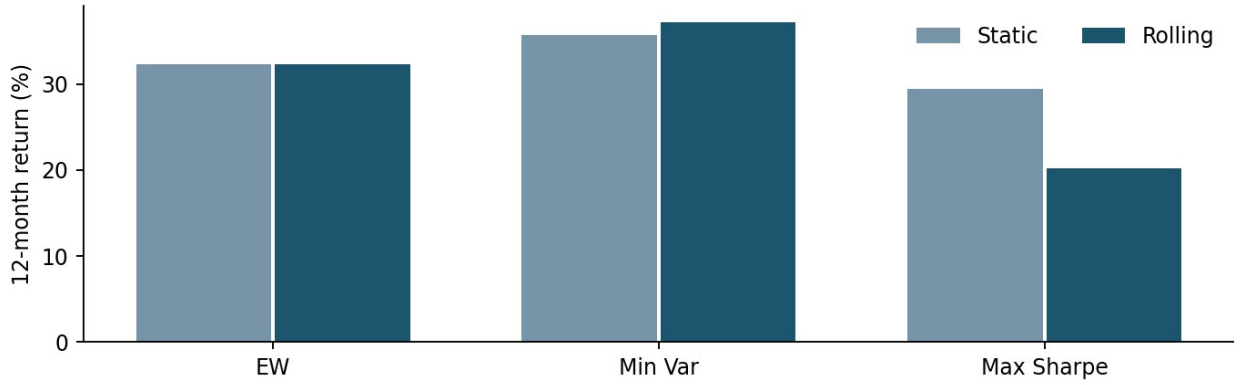
4 Rolling-window backtest: did reoptimization help?

Strategy	Static return	Rolling return	Change
Equal-Weighted	32.27%	32.27%	0.00 pp
Minimum Variance	35.70%	37.20%	+1.50 pp
Maximum Sharpe	29.40%	20.19%	−9.21 pp

The impact of reoptimization was asymmetric. Rolling Minimum Variance improved return by about 1.50 percentage points and reduced volatility from 13.44% to 12.86%; its maximum drawdown narrowed from 5.26% to 4.39%. Conversely, rolling Maximum Sharpe lost roughly 9.21 percentage points of return relative

to its static counterpart, while volatility increased from 15.93% to 16.70% and drawdown deepened to 10.86%.

This contrast is the report's principal analytical finding. Minimum Variance appears to have benefited from refreshing the covariance estimate as the test year progressed. Maximum Sharpe, which also relies on estimated expected returns, was less stable in realized performance. That interpretation is consistent with estimation sensitivity, although the workbook alone does not isolate the causal contribution of each input.



5 Trading costs, turnover and implementation

The rolling model applies a 0.10% cost to the amount traded, including the initial purchase. The workbook defines turnover as the sum of absolute changes between target and drifted weights; therefore the turnover figures are reported using that convention rather than a half-turnover measure. The modeled cost is a simplified execution assumption, not a full estimate of bid–ask spread, market impact or taxes.

Rolling strategy	Gross return	Net return	Cost drag	Avg. monthly turnover*
Equal-Weighted	32.27%	32.07%	0.20 pp	4.74%
Minimum Variance	37.20%	36.89%	0.31 pp	11.31%
Maximum Sharpe	20.19%	19.75%	0.44 pp	24.61%

*Average monthly turnover excludes the initial purchase. Cost drag is the gross-to-net return difference reported in the workbook.

Trading costs did not reverse the ranking: rolling Minimum Variance still produced the highest net return, while Equal-Weighted retained the lowest realized risk. Maximum Sharpe required substantially more trading, with average monthly turnover above twice that of rolling Minimum Variance, and still delivered the weakest rolling return. This weakens the case for frequent reoptimization of the Sharpe-maximizing allocation in this particular sample.

6 Sensitivity analysis and concentration

Risk-free assumption	Expected return	Expected volatility	Sharpe	WMT weight
2%	18.91%	15.60%	1.08	49.54%
4%	18.92%	15.61%	0.96	50.29%
6%	18.94%	15.63%	0.83	51.28%

Across the three risk-free-rate scenarios, the sensitivity worksheet keeps roughly half of the optimized Maximum Sharpe allocation in Walmart, while assigning no weight to Johnson & Johnson. The portfolio's composition changes only modestly as the assumed risk-free rate increases from 2% to 6%. That is evidence of limited sensitivity to this one input, not proof of overall robustness. The concentration itself is a material portfolio risk, and different expected-return assumptions could produce much larger weight changes.

7 Benchmark interpretation and limitations

The workbook currently reports 10.81% for SPY, but this is a scenario-dependent figure, not a verified market benchmark: BENCHMARK!B15 calculates October 2026 price from September 2026 using a manually entered −0.88% assumption in B25. Until the actual October 2026 adjusted closing price is sourced and the series independently checked, the precise benchmark return and outperformance margins must be treated as provisional. However, five actively selected stocks are not a risk-matched substitute for a diversified S&P 500 investment. The performance gap can reflect security selection, sector exposures and the specific market regime, not necessarily the optimization technique alone.

Several limitations matter. First, the 12-month out-of-sample period supplies only 12 realized monthly returns, so Sharpe ratios and drawdowns are imprecisely estimated. Second, all rolling windows overlap heavily, making their results dependent on much of the same historical information. Third, the workbook assumes a fixed 4% risk-free rate and simplified transaction costs. Fourth, the underlying stock and SPY price series should be independently checked for consistent dividend and corporate-action treatment. Finally, this is a five-security, long-only US equity exercise; the conclusions need not extend to other assets or periods.

8 Final investment recommendation

For a mandate that prioritizes risk-adjusted performance and lower realized downside risk within this five-stock equity universe, the Equal-Weighted portfolio is the preferred choice in this study. It returned 32.07% net of modeled trading costs, recorded a net Sharpe ratio of 2.59, and limited its maximum observed drawdown to 1.92%. It achieved these results without requiring estimates of relative expected returns to determine its weights. Its simple rules also make the strategy easy to explain, monitor and rebalance.

Rolling Minimum Variance is the strongest alternative for a return-seeking investor willing to accept a less defensive realized risk profile. It generated 36.89% net, approximately 4.82 percentage points more than Equal-Weighted, but with higher volatility and a drawdown more than twice as deep. The extra return is meaningful in the observed period; it should not be assumed to persist. Rolling Maximum Sharpe is not recommended on the current evidence, given its weaker net return, higher trading intensity and larger drawdown.

The next research priority is not another in-sample optimization. It is a longer, independent out-of-sample history with consistent total-return data, additional market regimes, explicit turnover and execution assumptions, and a wider asset universe. Until those tests are completed, the Equal-Weighted recommendation should be viewed as a sample-specific, risk-conscious conclusion rather than a forecast.

Source and calculation notes

Primary source: user-supplied APPLE(10).xlsx, especially the worksheets Monthly Returns, BACKTESTING, ROLLING BACKTEST, BENCHMARK, SENSITIVITY ANALYSIS and FINAL RECOMMENDATION. Figures are

rounded from workbook values. Returns labeled “annual” are total returns over a 12-month test. Sharpe ratios quoted in the comparison sections follow the workbook’s monthly-average annualization convention; gross and net results are not interchangeable. No external data or independent market-data validation was added. In particular, the October 2026 SPY return is assumed (−0.88%) in BENCHMARK!B25, so its 10.81% 12-month return is not verified. Historical stock and SPY adjusted-price sources and corporate-action treatment must be documented before public release. Publication condition: replace the SPY scenario assumption with a verifiable observed adjusted close, cite the data provider and retrieval date, and rerun benchmark-linked comparisons.